**20XT96 - MATHEMATICAL MODELLING LAB**

# GENOMEFOLIO — PACKAGE ABSTRACT

**Team members:**

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## Introduction

Financial return series are usually grouped with correlation. Correlation measures how much two series move together over a whole window, but it cannot recognise the same *pattern* when it appears a few days earlier in one stock and a few days later in another. This project instead treats each stock's return history as a **symbolic sequence**, groups stocks by **sequence similarity**, and forecasts each group's behaviour.

## Description

GenomeFolio implements a five-stage pipeline.

1. **Data** — daily adjusted prices for a fixed universe of US large-cap stocks are converted to simple daily returns on their common trading days.
2. **Symbolization** — each stock's returns over the trailing training window become a five-symbol alphabet: big-down, small-down, flat, small-up, big-up. Thresholds are per-stock quantiles of the training window, so one stock's volatility does not dominate the comparison.
3. **Alignment** — every pair of symbol strings is aligned with a banded **Needleman-Wunsch** global alignment using affine gaps and a graded substitution matrix. The band limits how far a pattern may shift, so unrelated days cannot be paired.
4. **Clustering** — the normalised alignment scores form a distance matrix, which is clustered with hierarchical average linkage. The number of clusters is chosen by silhouette under a constrained search with a minimum cluster size.
5. **Forecast** — each cluster's equal-weight composite return is forecast over the coming 21 trading days with ARIMA; a spline model is available as an alternative.

The pipeline stops at the forecast. It deliberately contains no null-hypothesis gate, no portfolio construction and no backtest.

**Datasets Used**

- **Primary** — 20 fixed cross-sector US large-caps, daily adjusted close from 2013 onward, downloaded once and cached locally.
- **Offline fixture** — a committed 10-stock sample covering five years, used by the tests and by the offline demo so the project runs without network access.

**Method and Implementation**

The implementation is pure Python 3.11 with NumPy, pandas, SciPy, scikit-learn and statsmodels; the banded alignment dynamic program is compiled with numba. Every parameter lives in `config/params.yaml`; nothing is hardcoded. Each stage uses only data up to the point being processed, so there is no look-ahead.

The distance between two symbol strings is the alignment score normalised by the two self-scores, `1 - score / sqrt(self_a * self_b)`. It is treated as a similarity measure, not as a mathematical metric: the code asserts symmetry and non-negativity, which is all average linkage requires.

**Expected Outcome**

The expected outcome is (a) interpretable groups of stocks whose return patterns are structurally similar and (b) a 21-day forecast for each group's composite. Because daily returns are only weakly predictable, the forecast is reported as a modest signal with its horizon, and limitations — survivorship, window sensitivity and weak predictability — are documented rather than hidden.